

conditional Generative Adversarial Networks (cGANs)

Objective: To understand and implement a Conditional Generative Adversarial Network (cGAN) using the Fashion-MNIST dataset. Compare the performance of cGAN with a standard GAN and analyze the impact of conditioning on generated image quality.

Problem Statement: Take the Fashion-MNIST dataset, develop a Conditional GAN that generates images conditioned on class labels. Compare results with a standard GAN and determine which model is better suited for controlled image generation.

Tasks:

- Download the Fashion-MNIST dataset (28x28 grayscale images of clothing categories).
- Normalize images and visualize sample images with labels.
- Implement a standard GAN (baseline).
- Implement a Conditional GAN (cGAN) using class labels.
- Modify Generator and Discriminator to include label conditioning.
- Train both GAN and cGAN models.
- Generate images using both models and compare outputs.
- Plot loss curves for Generator and Discriminator.
- Analyze differences between GAN and cGAN outputs.
- Determine which model is better suited and justify your answer.

Experiments must be performed:

- 1 Experiment by varying the size of the latent space and analyze its effect on image quality.
- 2 Experiment by varying the size of the class label embedding and analyze its impact.
- 3 Modify the architecture to concatenate class labels at different points in Generator/Discriminator and compare results.

Note: Students are advised to go through the class notes thoroughly before attempting the lab. You also can refer to online material. Conclusions for the experiments performed should be included. Make appropriate assumptions if needed and state them clearly.

YOU CAN ALSO TRY WITH OTHER TYPES OF GAN's (optional)

Deliverables:

- 1 Python code for GAN and cGAN implementations.
- 2 Plots of loss curves and generated images.
- 3 Comparative analysis report between GAN and cGAN.
- 4 Observations on additional experiments and conclusions.